Six finite worksheet implementations prepared

2026-10-18. Source-only work under ledger/handoffs/TASK-20261018-993753.yaml, authorized by DEC-20261018-59a33a and archived for implementation at commit 323663e23d292ebd42876332cd3dbc9448fdd3bc.

All six assigned contracts now have two separate scientific route modules, a direct native driver interface, an independently written finite oracle/checker, and a source-bound prospective trial layout. No frozen worksheet was evaluated. The implementation is ready for Coordinator source review and native-plan construction; it is not a ready/admitted scientific trial or a completed part of the original twenty-item audit.

Contract	Two measured routes	Independent checker
EXP-ECDLP-080496	z-first and x-first projective charts	Nonzero homogeneous vectors, quotient by scaling
EXP-ECDLP-4a3f03	Explicit beta walks and cubic-fibre partitions	Square-root table and equivalence closures
EXP-ECDLP-5a293e	x-outer and y-outer affine listings	Residue-indexed square-root census and separate map calculations
EXP-ECDLP-6491b4	Fraction arithmetic and Euclidean integer arithmetic	gcd-normalized independent cross products
EXP-ECDLP-65f903	Fraction inequalities and repeated integer products	Signed determinants and product lists
EXP-ECDLP-911c16	Fraction substitution and integer rational pairs	Independent coefficient cross products and gcd normalization

The exact sources are experiments/<EXP-ID>/implementation/{route_a.py,route_b.py,oracle.py,run.py,check.py}. Exact original inputs, controls, metrics, tails, stopping rules, criteria and out-of-scope limits are copied into preparation/<EXP-ID>/trial-layout.json. These are prospective layouts with placeholder IDs, not canonical trial plans. Each contract retains two separate RUN allocations, one per route; route b depends on route a and compares both complete measurements and canonical witnesses/point sets after checking its native successful dependency receipt and retained artifact hashes. Route a reports comparison pending rather than inventing agreement.

The shared worksheet_envelope.py handles metadata only. It binds actual supervisor launch, environment, command, plan, specification and source closure; writes outputs without overwrite; uses deterministic seed null; records real timing/resource boundaries; and sets the outer certificate kind to none. certificate.yaml is a finite transcript bound to the raw-result digest. It is not a formal proof, decomposition or discrete-log certificate. Checkers rederive measurements without importing either measured route. A false prediction is preserved separately from control validity and output integrity. Type-aware JSON comparisons reject Boolean/integer substitutions.

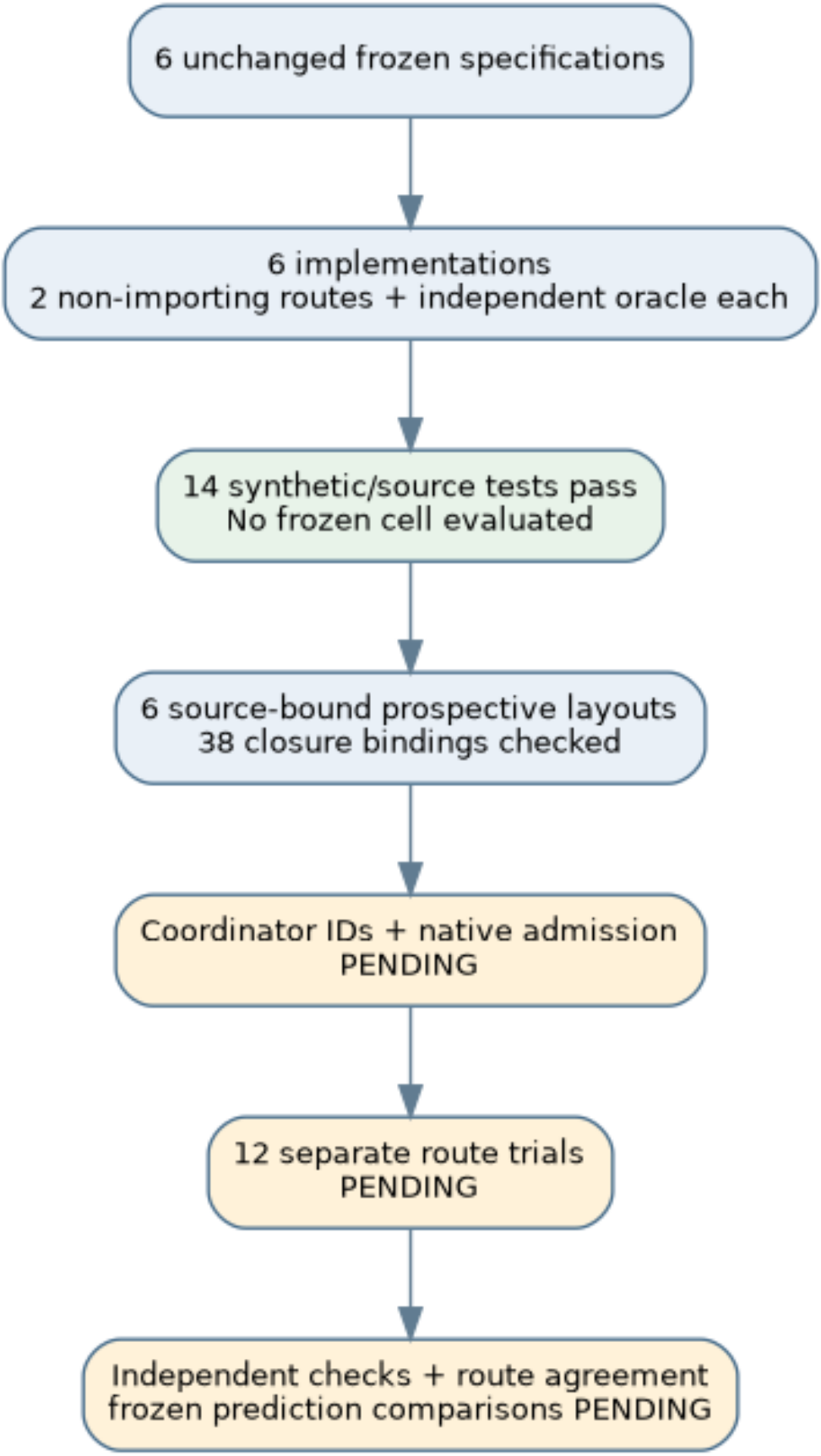
Fourteen software-only tests pass on distinct alternative fields/rationals and wholly mocked metadata fixtures. They cover three-way arithmetic/listing agreement, modular x+1, a coprime cancellation regression, false-prediction acceptance, the native minimum manifest gate, source binding, no overwrite, metadata/witness scope, Boolean scalar/resource tampering, changed previous artifacts and explicit cross-route witness equality. These tests ran in 0.261 seconds under a one-GiB address-space cap, one worker and the actual Python 3.12.14 runtime. This is a software-check duration, not a scientific measurement or performance comparison. The original complete frozen cells were not run. The inherited validator ResourceWarning is retained in stderr. See checks/software-controls-03.command.txt, .sources.sha256, .stdout, .stderr and .source-recheck.log.

All 32 final software-source hash entries pass. All 38 source bindings across six layouts pass, and the six frozen specifications match the Coordinator's protected hashes. implementation-receipt.json and checks/final-layout-validation.json contain exact paths, hashes, checks and unresolved obligations. Earlier smoke logs and prospective records before final witness-comparison hardening are retained. Internal read-only cross-agent source checks found no blocking issue before that final focused hardening, whose additional synthetic regression passes; final helper hash is 7841908dbed9caee0ff878aeb6b3c90e1327d80071e4f22061aab1dc6f8f12ad. This is not external or formal review.

Two bounded gaps remain explicit. In 4a3f03, denominator_mod has no formula in the supplied frozen specification/hypothesis; those two numbers are retained as declared inputs and are not claimed measured or verified. The native manifest gate accepts honest seed null, while the separate JSON schema still requires an integer seed. Neither gap is repaired by inventing a value. The absent recommended /usr/local/bin/python3 and lack of pinned Python 3.12.8 parity are also disclosed; prospective commands name the existing runtime.

The Coordinator still owns ID allocation, final source publication, artifact-complete plans/queues/handoffs, native preapproval/ready admission and subsequent scientific delegation. All twelve actual per-route worksheet runs, independent checks, final route agreements and frozen prediction comparisons remain unexecuted. No original specification, hypothesis, approval, old run, official scientific status, claim or leaderboard was changed. No commits, pushes or provider actions were performed by this implementation task.

No canonical scientific graph changes are implied by source preparation. The campaign pipeline source reports/ideas_execution_20261018/final-native-worksheets/pipeline.dot was inspected; its historical worksheet/manifest path remains historical evidence, while the diagram below separates this new source preparation from pending admission/execution. Unrelated isogeny/attack graphs have no new mathematical or measured datum to add.


6 unchanged frozen specifications
6 implementations
2 non-importing routes + independent oracle each
14 synthetic/source tests pass
No frozen cell evaluated
6 source-bound prospective layouts
38 closure bindings checked
Coordinator IDs + native admission
PENDING
12 separate route trials
PENDING
Independent checks + route agreement
frozen prediction comparisons PENDING